AEGIS STATION

Construction & Deployment Logistics

Modular. Redundant. Scaled for full Earth gravity. · 1g Configuration · November 2026 · Supersedes the April 2026 edition

Aegis Station is deployed in phases, ring by ring, with core operations starting long before full completion. This modular approach spreads cost, reduces risk, and enables early returns on investment. Each ring is a self-contained system — physically and operationally isolated from the others — allowing phased crew arrival, testing, and activation.

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1. Deployment Approach

Dry-orbit assembly and in-situ resource utilization as core principles.

Aegis Station is a modular, rotating habitat constructed in lunar orbit. Rather than launching a fully assembled station or supplying shielding water from Earth, the approach is built around dry-orbit assembly and in-situ resource utilization. Components arrive unshielded and are assembled in place; water for radiation shielding is sourced entirely from the Moon.

Water shielding is not a deliverable shipped from Earth. It is the first product of the lunar economy, extracted, processed, and delivered by the same logistics infrastructure that will sustain the station for decades.

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2. Assembly in Three Stages

Stage 1: Launch & Delivery

- Ring segments, hub modules, and dry structures launched on heavy-lift vehicles
- Orbital transfer between Earth and the Moon by the Earth–Aegis Long-Hauler fleet
- Each launch manifest is a discrete structural or systems package
- No water launched from Earth

Stage 2: Assembly in Lunar Orbit

- Assembly in a stable 100 km circular lunar orbit
- Axis-first: central hub modules are connected along the spine before ring construction begins
- Each ring is formed from 8 arc segments, robotically placed and locked into alignment
- Utility systems integrated at the segment level before ring closure
- The central hub is non-rotating; a ring spins up only after it is shielded and verified

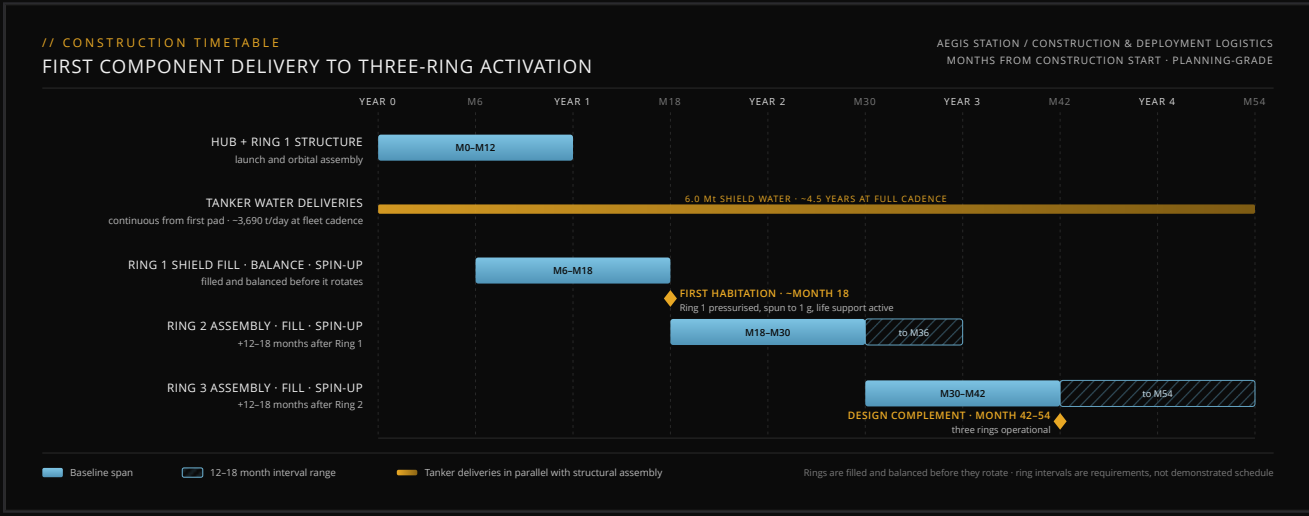
Stage 3: Shielding with Lunar Water

- Tanker operations begin in parallel with structural assembly
- A fleet of 82 tankers delivers 6.0 million metric tons of lunar water into the outer hull of the rings
- Rings are filled and balanced before they rotate; crew occupancy of a ring waits on its fill

3. Phased Ring Activation

Each ring is a self-contained, fault-isolated system with independent power, life support, and thermal control. This isolation supports phased crew arrival and systematic testing. Ring 1 serves as the pathfinder: it is spun up to 1g once its shield is filled and balanced and the ring is verified, hosts the initial operations and engineering crews, and validates all systems before Rings 2 and 3 follow.

Ring 1	Operational approximately 18 months from construction start; first pressurized, inhabited ring
Rings 2 & 3	Follow at 12–18 month intervals after Ring 1; each adds crew capacity, redundancy, and operational volume
Design complement	Three rings operational, month 42–54



Construction timetable, months from construction start. Planning-grade: ring intervals are requirements, not a demonstrated schedule.

4. Shielding Operations

Each ring receives water through five evenly spaced fill ports, supplied by cartridge-based delivery systems.

For the initial fill, the ring remains stationary. This simplifies mass distribution, avoids dynamic slosh complications, and allows for controlled pressurization of the shield layer. The ring structure is engineered to handle full shielding mass while static, and to spin up safely after shielding is complete.

Once filled, the ring is gradually brought up to its operational rotation rate to generate 1g at the outer floor. Top-offs and maintenance fills, which involve much smaller volumes, are made while the ring is spinning: sealed cartridges are carried through the hub/ring boundary on cargo pods and discharged inside the ring.

Per-ring shield mass	~2.01 million metric tons
Total shield mass	~6.0 million metric tons
Tanker payload	45 metric tons
Number of tankers	82
Daily delivery capacity	~3,690 metric tons / day
Fill time per ring	~18 months at fleet cadence
Full station fill time	~4.5 years

5. Materials & Equipment

What has to be made, moved, and installed.

What is built	Per ring	Three rings
Pressure hull and frames	~55,000 metric tons	~165,000
Hoop tension system	~92,000 metric tons	~276,000
Shield containment	~25,000 metric tons	~75,000
Decks, ~680,000 m² per ring	~34,000 metric tons	~102,000
Bulkheads, spokes, hub collar, margin	~14,000 metric tons	~42,000
Dry structure	**~220,000 metric tons**	**~660,000**
Arc segments	8, ~27,500 metric tons each	24
Shield water	~2.01 million metric tons	~6.0 million

Tanker fleet	82 tankers, 45 metric tons per sortie; ~134,000 cartridge loads over the fill
Pads and berths	11 surface pads and 11 station berths at the 3.2-hour turnaround ceiling; 15 recommended
Tanker engines	~5,400–10,700 over the campaign, as scheduled replacement items
Surface plant	Refinery of order 1.5 GW continuous; ~8,300 metric tons of water mined a day at fleet cadence
Earth-launched mass	~660,000 metric tons of ring structure; the hub is not yet stated
Not yet sized	Long-Hauler fleet, cartridge pool, interior outfitting

Structure masses are a concept-level allocation by function; materials and detailed design are left to the structural design. Program cost is not yet estimated.

6. Why This Works

- Modular deployment reduces critical path complexity
- Rings deliver incremental value before full buildout
- Shielding is integrated into the structure — no external tanks
- Long-hauler and tanker fleets operate in their optimal domains
- The 1g specification supports long-term health and crew retention

7. Fleet Architecture

Long-Hauler Fleet

Dry components from Earth to lunar orbit.

- Assembled and tested in LEO
- Fully reusable, modular vehicles
- Fleet size follows the structural delivery cadence (Section 10)
- Remains in service after construction for crew rotation and resupply

Lunar Tanker Fleet

Shield water from the lunar surface to the station.

- Assembled in lunar orbit
- Each tanker lifts 45 metric tons of water per trip
- Operates in closed loop: lunar surface → Aegis Station
- A fleet of 82 tankers fills the station in ~4.5 years

8. Logistics Flow

Earth Launches → LEO

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Long-Hauler Assembled & Deployed → LLO

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Aegis Station Assembled (dry) in LLO

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Tankers Assembled in LLO → Descend to Moon

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Tankers Lift Water → Return to Station in LLO

This path minimizes Earth lift cost, isolates risk domains, and maximizes throughput across the entire build sequence.

9. Modular Radial Modules for Gravity Simulation

Aegis Station enables precise gravity experimentation by installing modular segments along radial booms or structural spokes extending inward from the rings. These modules can be positioned at any desired radius from the central hub, enabling finely tuned simulation of partial gravity environments.

The station's rotation rate is fixed to generate 1g at the 397-meter floor. At this angular velocity (~0.157 rad/s, 1.50 RPM), specific gravity levels occur at the following radii:

Environment	Target gravity	Required radius
Earth (1g)	9.81 m/s²	397 m
Mars (0.38g)	3.73 m/s²	151 m
Moon (0.165g)	1.62 m/s²	66 m

- Human adaptation studies (cardiovascular, skeletal, muscular)
- Martian or lunar habitat prototyping
- Agricultural tests in simulated gravity
- Equipment validation in fractional G

10. Open Items

What does not yet close, carried as open.

Earth-launch cadence	The dry structure of one ring is ~220,000 metric tons. Delivering it within the first year of construction is about 600 metric tons a day to lunar orbit. That cadence is a requirement, not a demonstrated capability, and the Long-Hauler fleet is not yet sized for it.
Ring interval	At fleet cadence one ring's shield is about 18 months of deliveries. The 12-month end of the ring interval needs water delivered ahead of the ring or a higher cadence.
Structure mass	~220,000 metric tons per ring is a concept-level sizing for the spin and pressure loads with conventional materials. A bottom-up structural budget is not yet done, and the hub mass is not yet stated.
Cost	Not yet estimated. This edition states materials and equipment quantities, not prices.

11. Changes From the April 2026 Edition

Item	April 2026	This edition
Assembly location	Rings assembled in LEO and transferred by tug	Assembled in 100 km lunar orbit, axis-first, 8 arc segments per ring
Shield mass	~1.1 million metric tons per ring, ~3.3 million total	~2.01 million metric tons per ring, ~6.0 million total
Tanker fleet	45 tankers, ~2,025 metric tons / day	82 tankers, ~3,690 metric tons / day
Dry structure	~120,000 metric tons for the station	~220,000 metric tons per ring, ~660,000 for three rings
Cost breakdown	Stated by component	Removed; this edition gives materials and equipment quantities
Timeline	Illustrative, years 1–10	Months from construction start: Ring 1 at ~month 18, design complement at month 42–54
Top-offs	Synchronized cartridge vestibules on the spinning ring	Sealed cartridges through the hub/ring boundary on cargo pods
Gravity radii	1g at 400 m; Mars 152.3 m, Moon 68.1 m	1g at the 397 m floor; Mars 151 m, Moon 66 m